

END-OF-PROJECT REPORT

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Time series foundation models applied to Universal Domain Adaptation and Source Free Domain Adaptation

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1 - Introduction

Time series (TS) data is widely used to represent temporal trends in many application domains, including human activity recognition [18] and healthcare monitoring [19]. In many real-world scenarios, machine learning models trained on one dataset must generalize to data collected under different conditions. However, variations in sensors, environments, or subjects often lead to distribution shifts that significantly degrade model performance. Domain adaptation techniques aim to address this issue by transferring knowledge learned from a labeled source domain to an unlabeled target domain with a different data distribution.

In the context of time series, representation learning plays a key role in enabling effective transfer across domains. Recent advances in deep learning have introduced architectures specifically designed to capture temporal dependencies and complex patterns in TS data. Models such as TimesNet [5] and Fourier Neural Operator-based architectures [7] demonstrate that incorporating TS specific modeling strategies, such as frequency analysis or multi-period representations, can significantly improve performance across a variety of tasks. At the same time, the emergence of foundation models for time series [3,4] has introduced a new paradigm in which large-scale pretraining enables transferable representations that can be adapted to downstream tasks.

Despite these advances, models trained on large-scale datasets often face difficulties when deployed in new environments due to distribution shifts between training and test data. Unsupervised Domain Adaptation (UDA) attempts to address this challenge by learning domain-invariant feature representations using labeled source samples and unlabeled target samples [17,20]. However, most UDA methods assume that both domains share the same label set, which is often unrealistic in practical scenarios where new or unknown classes may appear in the target domain.

Universal Domain Adaptation (UniDA) extends this framework by relaxing this assumption and allowing the target domain to contain classes that are absent from the source dataset [1]. In this setting, models must simultaneously align samples belonging to shared classes while detecting target samples that correspond to previously unseen categories, often referred to as Out-of-Distribution (OOD) samples. Several UniDA methods have been proposed to address this problem, including adversarial approaches such as UAN [6] and optimal transport-based methods such as UniJDOT [8]. Unknown sample detection is typically performed using thresholding mechanisms based on metrics such as entropy [6-7], transport mass [8], or reconstruction error [21].

Although UniDA has been extensively studied in computer vision, research on domain adaptation for time series remains relatively limited. Recently, the UniDA benchmark for time series representation proposed in [1] introduced a systematic evaluation framework combining several adaptation algorithms and backbone architectures. However, the architectures evaluated in that benchmark are largely based on convolutional neural networks, and the impact of more recent TS specific architectures and foundation models has not yet been thoroughly investigated.

The objective of this work is to extend the UniDA benchmark introduced in [1] by incorporating modern time series architectures and foundation models as backbone feature extractors. In particular, we evaluate architectures specifically designed for time series representation learning, including TimesNet [5], as well as recent pretrained foundation models such as MOMENT [3] and Mantis [4]. These models have demonstrated strong performance in time series classification tasks, but their potential in the context of domain adaptation remains largely unexplored.

In addition to extending the benchmark with new backbone architectures, this study also initiates the exploration of Source-Free Universal Domain Adaptation (SF-UniDA) for time series. In this setting, the original source data are no longer available during the adaptation phase, which reflects practical constraints such as privacy restrictions or storage limitations. While SF-UniDA has been studied in other domains, its application to time series remains relatively underexplored.

Through this work, we aim to study the suitability of modern time series architectures and foundation models in the UniDA setting and to complement the existing benchmark with additional backbone architectures. By expanding the range of models evaluated within the framework proposed in [1], This study provides further insight into the role of TS-specific representations and foundation models under domain shift. Additionally, this study aims to make the first steps into transforming the UniDA benchmark into a SF-UniDA benchmark on TS.

The remainder of this paper is organized as follows. Section 2 presents the background on time series representation learning, Universal Domain Adaptation, and Source-Free Domain Adaptation. Section 3 introduces the extended benchmark and the backbone architectures considered in this work. Section 4 describes the experimental setup, including datasets, evaluation protocols, and training settings. Section 5 presents and discusses the experimental results. Finally, Section 6 outlines future research directions, and Section 7 concludes the paper.

2 - Background

2.1 - Time Series in Domain Adaptation

A time series (TS) can be defined as an ordered sequence of observations collected over time at regular intervals. Formally, a time series x can be represented as $x = \{x_1, x_2, \dots, x_T\}$, where T denotes the number of observations $xt \in R^D$ of D measured features. Time series often exhibit temporal patterns such as long-term trends, periodic or seasonal components, and irregular fluctuations caused by external factors or noise.

In machine learning, time series are commonly processed through representation learning, where a model f transforms the raw sequence x into a latent representation z . This transformation can be expressed as $f: x \rightarrow z$, where $z = \{z_1, z_2, \dots, z_R\}$ and each latent vector $z_r \in R^K$. The resulting representation typically compresses or reorganizes the original information so that $R \leq T$ and $K \leq D$, enabling models to capture the most informative temporal structures.

Working with time series in the context of domain adaptation introduces challenges that are less prominent in fields such as computer vision. Unlike image datasets, which benefit from large-scale datasets and well-established baseline models, the literature on domain adaptation for time series remains comparatively limited [13,14]. One reason is the difficulty of data annotation: labeling temporal signals often requires expert knowledge to interpret patterns or events within the sequence. This process can be time-consuming and costly, making the creation of large annotated datasets difficult. As a result, the number of publicly available datasets for time series domain adaptation is relatively small.

2.2 - Universal Domain Adaptation

Machine learning models might be trained on data collected in one environment but later deployed in another where the data distribution differs. Domain adaptation addresses this problem by transferring knowledge from a labeled source domain to a target domain for which labels are unavailable or scarce [2]. Let the source dataset be defined as $D_s = \{(x^s, y^s)\}$, where x^s denotes the input sample and y^s its corresponding label, following a joint distribution $P^s(x^s, y^s)$. The target dataset is defined as $D_t = \{x^t\}$ containing only unlabeled samples, drawn from a different distribution $P^t(x^t, y^t)$, with $P^s \neq P^t$.

In domain adaptation, it is predefined that both domains share the same label set. This approach simplifies the task, but it is not realistic because assumptions about classes must not be made about the unlabeled target set. This leads to UniDA, where the set of labels present in the source and target domains are not identical, to avoid any assumption on the label set distribution.. The relationship between the two domains can therefore be described through three subsets of labels. The common label set $Y_C = Y_s \cap Y_t$ contains the categories that appear in both domains. Samples belonging to these shared classes should be correctly aligned across domains so that the model can generalize the learned decision boundaries. The private source label set $\bar{Y}^s = Y_s \setminus Y_t$ includes classes that exist only in the source domain. Since these categories are absent from the target data, aligning them with target samples would lead to incorrect predictions and must therefore be avoided. Finally, the private target label set $\bar{Y}^t = Y_t \setminus Y_s$ contains categories that appear exclusively in the target domain. Because the model has never encountered these classes during training, they must be treated as unknown categories and identified as such during inference.

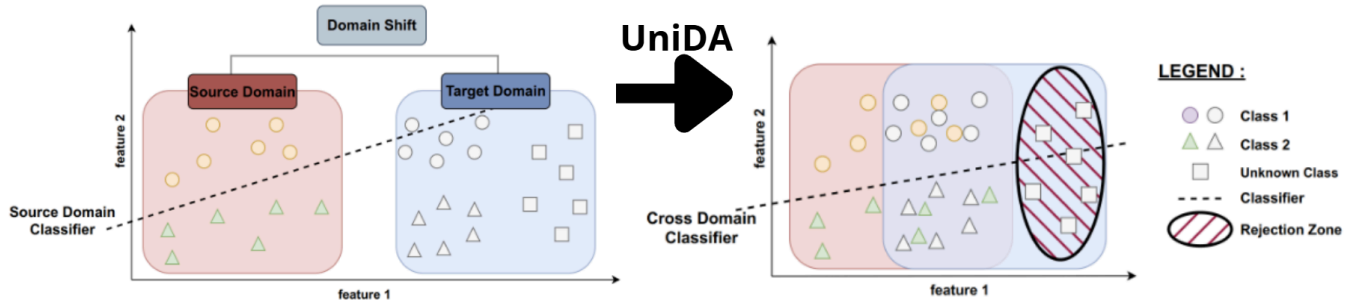


Figure 1 - Illustration of Universal Domain Adaptation (UniDA). In the left image, a classifier trained on the source domain fails to generalize to the target domain. In the right image, UniDA aligns shared classes across domains while detecting unknown target samples, assigning them to a rejection zone [1].

Given this configuration, the learning problem involves two main objectives that can be identified in Figure 1. First, the model must learn to align the representations of samples belonging to the shared classes so that the knowledge learned from the labeled source data can be transferred to the target domain. Second, the model must be capable of detecting samples from unknown target classes, preventing them from being incorrectly classified into known categories.

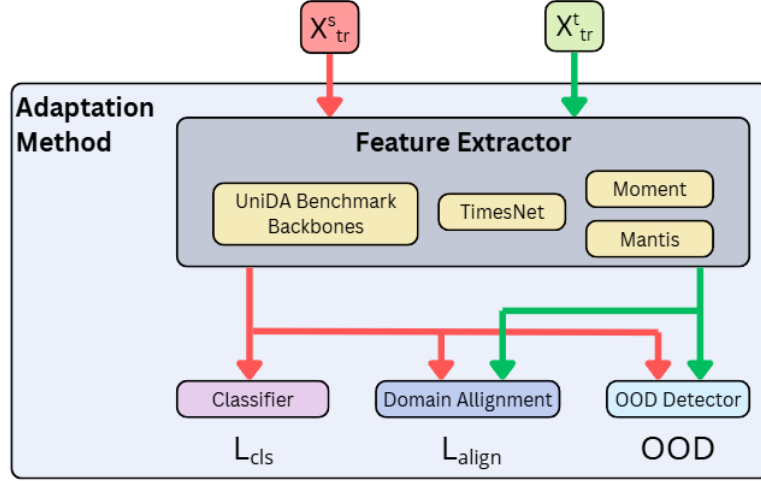


Figure 2 - For UniDA, source and target data are used simultaneously to train a feature extractor with a classification and an alignment loss, while also detecting the points out of distribution.

Approaches typically address this problem using two main components: a feature extractor g , which maps the input data to a latent representation, and a classifier f , which predicts class labels from these features. The model is trained using a classification loss, most commonly cross-entropy, computed on labeled source samples to learn the main task. In addition, an alignment loss is used to reduce the discrepancy between source and target feature distributions [12]. Feature alignment can be decomposed into two families: discrepancy-based [8, 17], or Adversarial [6]. To handle target-specific classes, a mechanism is usually incorporated to detect samples that do not belong to the shared categories, often relying on uncertainty measures [8] or entropy thresholds [6,7] in the learned feature space [1]. This structure can be seen in Figure 2.

2.3 - Source Free Domain Adaptation

In some practical situations, access to the original source dataset is not possible during the adaptation stage. This setting arises for several reasons, such as data privacy restrictions, intellectual property constraints, or storage limitations. In such cases, only a pre-trained model trained on the source domain is available, while the raw source samples cannot be reused. The goal is therefore to adapt this model to an unlabeled target domain without revisiting the original training data.

In SF-UniDA, a model composed of a feature extractor g and a classifier f is trained on D^s using a standard supervised objective, typically a cross-entropy loss. After training, the source data D^s are no longer accessible. The adaptation phase is then performed using only the trained model parameters and D^t . The challenge is therefore to modify or refine the model so that it performs well on the target domain despite the absence of source samples during adaptation.

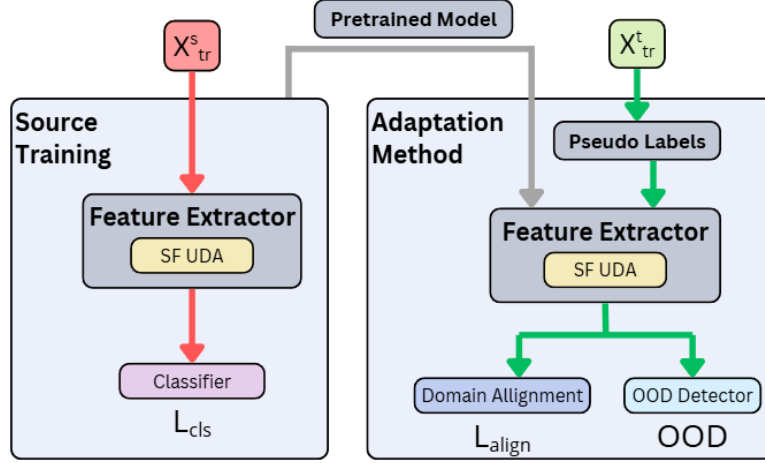


Figure 3 - For SF-UniDA, source data is used to pretrain a model with a classification loss, while the target data goes to a pseudo labelling process to create what may be the private source classes and then the model is trained to align both domains and detect OOD points.

This setting shares similarities with the framework described in the previous section, as the objective remains to address the distribution discrepancy between source and target domains. However, unlike traditional approaches that rely on explicit alignment between source and target features, the absence of source data requires alternative strategies like the ones in [10, 11, 16]. Instead of directly learning from samples from both domains, adaptation methods must rely on the knowledge encoded in the trained source model. For instance, pseudo-labeling techniques can be used to assign tentative labels to target samples based on the model’s predictions, which are then used to iteratively refine the decision boundaries. For example, [15] has a self-supervised approach, while [16] does the pseudo labeling through feature decomposition without requiring the target label set. The scheme is shown in Figure 3.

3 - Extended UniDA Benchmark

This work builds upon the benchmark introduced in [1]. In that study, several domain adaptation algorithms were evaluated on time series classification tasks under the universal adaptation setting. Here, we consider the best-performing algorithms and backbone architectures reported in the reference work as baseline methods. These models serve as a comparison point for the foundation models for time series analyzed in this paper.

The benchmark relies on a modular framework in which adaptation algorithms are paired with feature extraction backbones. The algorithms define how the model performs cross-domain alignment and unknown sample detection, while the backbone architecture determines how temporal features are extracted from the input time series. In the following subsections, we briefly describe the main algorithms considered in the benchmark as well as the backbone architecture that achieved the strongest empirical performance. The benchmark is composed mainly of CNN-based feature extractor, here we provide TS-oriented feature extractors to the analysis. Among the studied models, we considered TimesNet, MOMENT and Mantis, this extension of UniDABench is shown in Figure 2.

3.1 - Algorithms

The algorithms present in the original benchmark proved to be suited to UniDA on TS, despite being designed for domain shift on image tasks. This study brings three of the best performing algorithms from the benchmark (UAN, DANCE and UniJDOT) and complements the study with SF-UAN, a contribution to the study of the SF-UniDA task. Those algorithms are described as follows:

- UAN [6]: The Universal Adaptation Network proposes a strategy to quantify the transferability of individual samples during training, integrating both the domain similarity and the prediction uncertainty of each sample into a weighting mechanism. UAN uses an adversarial domain discriminator to learn domain-invariant features and perform alignment between source and target samples. At the same time, potential unknown target samples are detected through a rejection mechanism based on the entropy of a domain classifier, allowing the model to separate shared classes from unseen categories.
- DANCE [7]: Domain Adaptive Neighborhood Clustering via Entropy Optimization approaches the problem from a different perspective by exploiting the intrinsic structure of the target domain. Instead of relying exclusively on source supervision, the method introduces a self-supervised mechanism called neighborhood clustering, which encourages target samples with similar representations to form clusters in the feature space. During training, these clusters are gradually aligned with the source domain while preserving discriminative boundaries between classes. The method also introduces an entropy separation loss, which allows the model to either associate a target sample with a known source class or reject it as belonging to an unknown category. Unknown samples are detected by applying a threshold on an entropy-based margin computed from the classifier outputs.
- UniJDOT [8]: This is a method specifically designed for time series classification that extends optimal transport-based domain adaptation approaches to the universal setting. The method relies on Optimal Transport (OT) to align source and target samples in a joint feature-label space. To improve robustness, UniJDOT introduces a joint decision space in which classifier predictions are adjusted using distance-based probability vectors derived from the feature space. The approach also employs an automatic thresholding strategy to pseudo-label target samples as either belonging to the shared classes or to the unknown set. This separation enables the model to align common classes through optimal transport while simultaneously isolating samples that likely belong to unseen categories.
- SF-UAN: This is an initial model for the SF-UniDA task constructed based on the UAN algorithm for UniDA, therefore it follows the training process described in Figure 3. It was developed in this paper as an initial step of transforming the main UniDA benchmark's architecture into SF-UniDA. The adversarial domain discriminator from the base algorithm is still present but utilizing only the target domain data, while the model works in a new set of pseudo labels in order to identify the potential unknown classes from the unseen dataset. This pseudo labeling is made through a self-supervised labelling from a few samples of the target dataset, as it is done in [15].

3.2 - Backbone

3.2.1 - FNO

Among the architectures evaluated in the benchmark, the Fourier Neural Operator–inspired backbone (FNO) demonstrated the best overall performance in the results from [1]. This architecture integrates both time-domain and frequency-domain representations of the input signal. The process begins by applying the Fast Fourier Transform (FFT) to project the time series into the frequency domain. Convolution operations are then performed on these frequency coefficients to capture global temporal dependencies. Finally, the representation is transformed back to the original space using the inverse FFT (iFFT) [7].

In practice, the model often combines a conventional convolutional neural network (CNN) operating in the original time domain with the frequency-domain features produced by the FNO layer. The resulting representation merges information from both domains.

3.2.2 - TimesNet

TimesNet is a deep learning architecture designed to model complex temporal variations in time series data, so it has been chosen as a potential backbone for comparison to the ones in the benchmark since it is TS oriented. A key observation motivating TimesNet is that many TS exhibit multi-periodicity, meaning that temporal patterns occur at multiple overlapping periodic scales. Traditional models operating directly on the 1D sequence often struggle to capture these interactions.

To address this limitation, TimesNet transforms the original 1D time series into 2D representations based on multiple period sizes, enabling the model to simultaneously capture intra-period and inter-period variations [5]. In this formulation, columns represent variations within a period while rows capture variations across different periods.

The model introduces a modular component called TimesBlock, which first identifies dominant periods using frequency domain analysis and then reshapes the sequence accordingly. Each reshaped representation is processed using inception convolutional blocks that aggregate temporal patterns at multiple scales. Then the final result is once again combined into a single TS.

3.3 - Foundation Models for Time Series

To enrich the UniDA benchmark, three new state-of-the-art backbones were evaluated. Those models were pretrained on large-scale TS datasets, which may lead to promising results since they differ from the backbones analysed so far that require training from the start. MOMENT and Mantis were evaluated on their respective papers on a regular classification task without domain shift. This study aims to analyse whether foundation models are suited for the domain adaptation task in a universal setting.

3.3.1 - MOMENT

MOMENT is a family of large-scale transformer-based time series foundation models designed to enable general-purpose temporal representation learning. It leverages large-scale

pre-training to learn transferable time series representations that can be adapted to a wide range of tasks [3].

As mentioned previously, a major challenge in training large time-series models is the lack of a large-scale dataset. To address this issue, for the training of MOMENT, the Time Series Pile has been introduced, a collection of publicly available datasets spanning multiple domains. This dataset enables multi-domain pretraining, allowing the model to capture diverse temporal structures and patterns.

Architecturally, MOMENT adopts a transformer-based design operating on time-series patches, where subsequences of the signal are treated as tokens. This patch-based representation improves efficiency and enables the model to capture long-range dependencies in temporal data. During pre-training, the model employs a masked reconstruction objective, where portions of the input time series are masked and the model learns to reconstruct them. This self-supervised learning strategy encourages the model to learn meaningful temporal representations without requiring labeled data.

3.3.2 - Mantis

Different from MOMENT, Mantis is a time-series foundation model designed specifically for time series classification. It follows a representation learning paradigm in which an encoder learns to map a time series sequence into a discriminative embedding space, which can then be used for downstream classification tasks [4].

The architecture combines a token generation module with a Vision Transformer (ViT) backbone. The input time series is first divided into patches that are converted into tokens, which are then processed by the transformer encoder to produce a latent representation. The model is pre-trained using large collections of unlabeled time series data, allowing it to learn generic temporal representations that transfer well to new datasets.

4 - Experiments

4.1 - Model selection

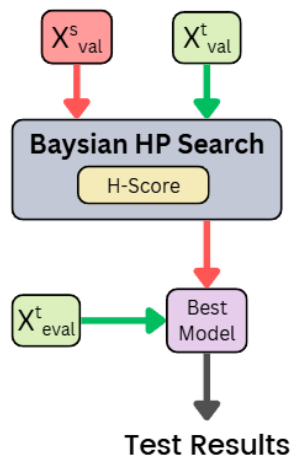


Figure 4 - The best HP are defined through the evaluation of the H-score across multiple scenarios on the validation set, then the final result is determined from the evaluation set.

Domain adaptation datasets often contain several domains, each corresponding to data collected under different conditions, environments, or subjects. These domains allow models to be evaluated across multiple adaptation scenarios. In practice, each domain can be used either as a source or as a target. By pairing domains together, different adaptation scenarios can be constructed, where one domain provides the training supervision D^S and another serves as the unlabeled target on which the model must generalize D^T . To ensure a fair evaluation, the set of all possible domain pairs is divided into two groups: a validation set S^{val} , used for hyperparameter selection, and an evaluation set S^{eval} , used to report the final performance of the models.

To measure the effectiveness of adaptation methods in this setting, models are evaluated using the H-score, calculated as $(2A_C A_U)/(A_C + A_U)$, where A_C is the accuracy on common classes and A_U on unknown classes of the target domain. This metric combines two complementary objectives: the ability of the model to correctly classify samples belonging to shared classes between domains, and its ability to correctly identify unknown classes that appear only in the target domain. This formulation penalizes models that perform well on only one of these objectives while failing on the other.

For each adaptation scenario, the target dataset is further divided into two subsets. One subset is used during training as unlabeled target data, allowing the model to adapt to the target distribution without accessing ground-truth labels. The second subset is reserved for evaluation, where labels are used only to compute the final H-score after training.

4.2 - Dataset

The experiments are conducted using the Human Activity Recognition (HAR) dataset, a widely used benchmark for time series classification tasks. This dataset contains recordings of daily human activities collected from wearable sensors attached to participants. The signals correspond to motion measurements obtained from smartphone sensors, including accelerometers and gyroscopes, which capture body movements during different physical activities.

The dataset contains six activity classes: walking, walking upstairs, walking downstairs, sitting, standing, and lying. Sensor signals are segmented into fixed-length windows of 128 time steps, with each window representing a short temporal segment of an activity. Each sample consists of nine channels, corresponding to different inertial signals derived from the raw sensor measurements. These channels include three-axis body acceleration, three-axis gyroscope measurements, and three-axis total acceleration components. The dataset is composed of 30 domains, which correspond to different people who registered the data while performing the simple activities. Thus, the domain adaptation consists of training a model with a participant's data in such a way that it also performs well on other participants.

HAR provides a rich multivariate time series representation of human motion and has been widely used in activity recognition research. In this study, this was the only dataset analysed due to limited resources, but many other datasets, such as WISDM, SSC and MFD are commonly used for domain adaptation and can improve this research [2].

4.3 - Experimental Settings

For the universal domain adaptation task, each model was trained for 20 epochs. Regarding model architectures, the FNO backbone integrates a Fourier Neural Operator layer whose output is concatenated with convolutional neural network (CNN) features extracted in the time domain, following the same approach as [1].

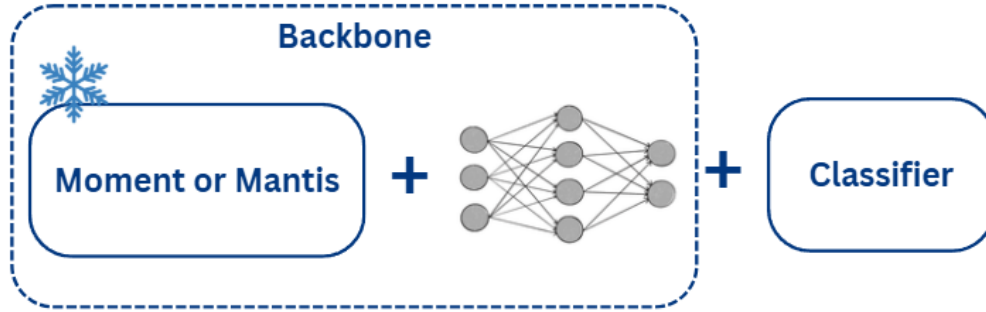


Figure 5 - The backbone from foundation models are composed of the pre-trained model with gradients frozen and an additional MLP that is trained over the epochs.

The TimesNet architecture consists of a single TimesBlock followed by a fully connected neural network layer for classification. For MOMENT and Mantis, the pretrained foundation models are used as fixed feature extractors with frozen gradients, and an MLP head is trained on top of the extracted representations, shown in Figure 5.

Hyperparameter selection was performed using the Weights & Biases platform. For each model, 20 hyperparameter sweeps were executed over 10 different adaptation scenarios, with a single run per configuration. While this search strategy is relatively limited compared to large-scale hyperparameter optimization procedures, it was constrained by the computational resources available through the free GPU allocation of Google Colab. After selecting the best-performing hyperparameters on the validation scenarios, the final models were trained with five independent runs across ten different scenarios in order to obtain more stable and reliable performance estimates.

In the case of the source-free domain adaptation task, models were first pretrained on the source domain for 20 epochs, after which the pretrained model was further trained on the target domain for an additional 20 epochs without access to source data.

5 - Results and Discussion

HAR	Mantis			Moment			TimesNet			FNO		
	UAN	DANCE	UniJDOT	UAN	DANCE	UniJDOT	UAN	DANCE	UniJDOT	UAN	DANCE	UniJDOT
12 → 16	74,9	42,6	49,3	38,3	31,9	31,3	50,8	31,5	50,2	49,6	46,3	46,6
13 → 3	88,8	40,8	81,3	33,1	40,2	37,2	66,8	57,7	72,4	71,2	70,1	72,4
15 → 21	62,6	35,3	59,1	40,2	26,6	34,6	50,3	51,2	61,6	66,5	54,8	73,5
17 → 29	74,2	31,1	47,5	37,2	31,6	33,9	46,1	35,6	56,4	51,2	53,3	58,0
1 → 14	99,1	33,6	69,6	32,9	28,9	25,5	59,6	67,1	75,6	69,2	72,1	94,0
22 → 4	70,1	32,1	57,0	34,4	26,5	27,9	39,4	41,8	51,4	58,7	37,0	68,9
24 → 8	57,7	45,9	52,4	41,5	38,3	30,5	47,4	44,6	59,8	49,4	42,8	52,6
30 → 20	59,0	26,9	56,0	37,6	26,7	17,5	44,7	35,0	50,2	38,7	39,6	42,1
6 → 23	56,5	38,6	56,5	47,8	37,5	38,1	33,9	37,9	58,1	51,9	64,3	60,7
9 → 18	66,2	31,9	59,2	32,4	32,6	39,0	39,8	34,2	59,5	71,6	44,6	58,2
Mean	70,9	35,9	58,8	37,6	32,1	31,5	47,9	43,7	59,5	57,8	52,5	62,7
Std	14,0	5,9	10,0	4,8	5,1	6,6	9,8	11,6	8,7	11,3	12,7	15,1

Table 1 - H-score for each algorithm+backbone combination over 10 different scenarios.

Table 1 reports the mean H-score obtained by each algorithm–backbone combination across ten adaptation scenarios, averaged over five runs using the selected hyperparameters.

Overall, the best performance was obtained by the UAN + Mantis configuration, which outperformed all models reported in the original benchmark [1], but the backbone truly underperforms when used in the DANCE algorithm. This result highlights the strong interaction between the algorithm and the feature extraction backbone for domain adaptation. Additional tests on other datasets are required to better understand this behavior.

Comparing the two pretrained datasets, we see that Mantis performed better overall, while MOMENT results are unsatisfactory. The superiority of Mantis performance over MOMENT was also observed on [4] on a regular classification task (without domain shift) with fine-tuning of both models. It is important to add that Mantis presented a much lighter and faster model, achieving its purpose of being lightweight.

TimesNet achieved results equivalent to most of the models in the original benchmark, but it is important to clarify that the combination of TimesNet and DANCE required some adjustments to work. DANCE uses a fixed entropy threshold ρ in order to reject OOD points ($\rho = 0.5 \cdot \log(NbClasses)$), but TimesNet entropy results were all too low, even for samples belonging to private target classes. As a result, the fixed entropy threshold used by DANCE failed to reject any out-of-distribution samples, leading to an H-score of zero. That is, TimesNet was always too confident in its predictions to identify private classes. To address this issue, the rejection threshold was modified to scale with the maximum observed entropy within the batch for TimesNet+Dance according to a new hyperparameter. This new input ϵ is the percentile of all entropy to be considered as a common class ($\rho = percentile(Entropy, \epsilon)$), allowing unknown samples to be detected as the entropy values bigger than the calculated ρ .

HAR	FNO Limited			FNO Benchmark		
	UAN	DANCE	UniJDOT	UAN	DANCE	UniJDOT
12 → 16	49,6	46,3	46,6	56,8	49,9	50,6
13 → 3	71,2	70,1	72,4	67,3	71,4	76,3
15 → 21	66,5	54,8	73,5	78,4	66,6	81,5
17 → 29	51,2	53,3	58,0	63,5	65,3	78,3
1 → 14	69,2	72,1	94,0	63,8	64,8	39,2
22 → 4	58,7	37,0	68,9	63,0	41,2	72,4
24 → 8	49,4	42,8	52,6	50,7	38,6	47,0
30 → 20	38,7	39,6	42,1	47,2	49,4	53,4
6 → 23	51,9	64,3	60,7	66,6	68,5	78,7
9 → 18	71,6	44,6	58,2	65,4	25,4	68,6
Mean	57,8	52,5	62,7	62,3	54,1	64,6
Std	11,3	12,7	15,1	8,9	15,5	15,5

Table 2 - H-score for FNO with the limited resources and benchmark FNO over 10 different scenarios.

In Table 2, FNO Limited’s results were obtained from a model obtained from the limited experiment settings from this paper, while FNO Benchmark are the benchmark’s final results from [1], where the hyperparameter search was much more extensive. Since the benchmark results rely on a significantly larger hyperparameter search space, it is possible that the performance of the evaluated foundation models could be improved through a more extensive model selection procedure.

HAR	Mantis	FNO
	SF-UAN	SF-UAN
12 → 16	48,7	37,6
13 → 3	79,0	88,8
15 → 21	36,0	0,6
17 → 29	65,4	52,7
1 → 14	21,5	40,5
22 → 4	34,1	40,5
24 → 8	74,8	92,1
30 → 20	93,2	60,6
6 → 23	52,0	70,0
9 → 18	59,0	70,9
Mean	56,4	55,4
Std	22,4	27,3

Table 3 - H-score for Mantis and FNO for SF-UniDA task..

Table 3 brings the results for Mantis and FNO, the best performing architectures in UniDA, but this time applied in an SF-UniDA task with SF-UAN. We can see that they both obtained very similar results that are lower than the ones obtained on the UniDA task. This exemplifies that the absence of the source data makes the domain adaptation more difficult to be accomplished.

Finally, it is worth mentioning that the standard deviation across scenarios is relatively high for all methods. This variability is likely related to the limited size of the HAR dataset, which contains relatively few samples per domain.

6 - Future Works

First, a deeper hyperparameter search should be conducted for the newly introduced backbone architectures. Due to computational constraints, the current experiments relied on a relatively limited hyperparameter exploration. A more extensive search could reveal stronger performance for the

evaluated models and provide a more accurate comparison with the original benchmark results. Second, the benchmark itself should continue to evolve by incorporating newly proposed state-of-the-art algorithms and backbone architectures.

Finally, further work is required to properly investigate Source-Free Domain Adaptation for time series. This includes implementing additional SF-UniDA algorithms, such as GLC++ [11] and MAPU [10], and establishing a dedicated benchmark setting for this problem.

7 - Conclusion

This work extended the Universal Domain Adaptation benchmark for time series proposed in [1] by evaluating recent time-series-specific architectures and foundation models as backbone feature extractors. The experiments showed that the choice of backbone significantly influences adaptation performance, with the Mantis backbone combined with UAN achieving the best results among the evaluated configurations.

The results also highlighted the importance of compatibility between adaptation algorithms and backbone architectures, as some combinations led to significant performance variations. Overall, this study provides an initial evaluation of modern time series architectures within the UniDA framework and contributes to extending the benchmark with new backbone models.

Additionally, preliminary experiments on Source-Free Domain Adaptation with the original algorithm SF-UAN demonstrated the increased difficulty of the task when source data are not available during adaptation. The SF-UAN algorithm was constructed based on other state-of-the-art algorithms from TS UniDA and computer vision SF-UniDA and is a naive approach to the problem, but it facilitates the process of adapting new algorithms to a future SF-UniDA benchmark.

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